AIDE

LLM Rephrasing Stage — Prompt & Model Evaluation Report

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Scope: selecting the prompt and open-weight model that turn a raw speech-to-text transcript into a short, de-identified action item for the ward dashboard.

Headline decision: lock **qwen3:4b** (Apache 2.0) with prompt **v2**. 83.6% exact match, 90.2% semantic match, and **zero PII leaks** across 225 test transcripts including a 100-row split with names, ages and towns injected.

This report covers three evaluation runs: a cloud baseline (prompt v1), a prompt revision (v2), and the deciding run — self-hosted small models at production quantisation (v2).

1. Task and method

The rephrasing stage is a narrow text-transformation problem, not open-ended reasoning. Input: one patient's spoken request (already through speech-to-text). Output: a 2–5 word Title Case action item ending in a fixed category noun (request, assistance, adjustment, question, update), with every name, age and location removed.

Approach, in order of preference: (1) prompt engineering on an existing model; (2) fine-tuning a small model later, once real labelled pilot data exists; (3) training from scratch — ruled out. This report is entirely stage (1).

A reusable evaluation harness runs every transcript through a given prompt + model, then scores the output three ways and logs every mismatch. The same harness targets cloud APIs (NVIDIA NIM, Groq) and a local Ollama server, so results are directly comparable.

2. Dataset

225 transcripts, `eval_set_225.csv`:

Split	Rows	What it tests
general	100	clean requests → correct short label
personal_data_injected	100	same requests + name / age / town → label must strip all of it
realistic_messy	25	disfluent, run-on, ASR-style input

Labels follow a closed house style (see `LABEL_CONVENTION.md`). The three source datasets were checked; only a 25-row auxiliary file was off-convention (3 labels corrected).

3. Metrics

Metric	Definition
exact	normalised output == expected label, case-sensitive (catches format drift)
semantic	same category noun (or topic) and ≥ 50% content-word overlap after singularising
PII leak	output contains a digit, or a patient name / town from the transcript not in the label. HARD GATE — any leak disqualifies a m

All three reported separately per split, never blended.

4. Prompt: v1 → v2

v1 stated the rules and gave four examples. It achieved zero PII leaks but low exact match: models understood every request but did not match AIDE's specific vocabulary — over-capitalising (“Water Request”), saying “Phone request” where the label wants “Phone retrieval assistance”, singular vs plural, over-specifying, and firing “Unclear request” on real-but-vague needs (“I'm not feeling well”).

v2 adds: capitalise the first word only; an explicit retrieval-vs-request rule; a tightened “Unclear request”; a list of ~65 preferred labels; and ~24 few-shot examples built directly from v1's failure cases, deliberately mixing clean and PII-injected inputs so de-identification is demonstrated, not just described.

On a 50-row cloud smoke test (qwen3.8-27b, general split): exact 60% → **92%**, semantic 86% → **100%**.

5. Results

5.1 Run 1 — prompt v1, cloud (Groq), 225 rows

Model	Exact	Semantic	general	personal	messy	PII leak
qwen3.8-27b	62.2%	84.4%	81%	89%	80%	0
gpt-oss-120b	11.1%	83.1%	82%	86%	76%	0
gpt-oss-20b	47.6%	78.7%	77%	83%	68%	0

Every model, zero leaks. gpt-oss-120b's 11% exact is pure capitalisation. Per-row cloud outputs were not retained; Run 3 (below) is the deciding run.

5.2 Run 3 — prompt v2, self-hosted (Ollama, CPU, Q4), 225 rows

This is the deciding run: the pilot forbids external API calls in the request path (data residency), so the model must be self-hosted. These are the real quantised models on a real box.

Model	Params	Licence	Exact	Semantic	personal	PII leak	p50
qwen3:4b	4B	Apache 2.0	83.6%	90.2%	94%	0	2.4s
gemma3:4b	4B	Gemma	76.6%	84.4%	(finishing)	0	1.9s
llama3.2:3b	3B	Llama	74.2%	85.8%	90%	0	1.4s
phi4-mini	3.8B	MIT	74.2%	84.4%	89%	0	1.6s
qwen3:1.7b	1.7B	Apache 2.0	70.7%	89.8%	91%	0	0.9s
granite3.1-moe:3b	3B	Apache 2.0	(finishing)				

Every model: **zero PII leaks** — the v2 de-identification instruction holds across the board. qwen3:4b wins by 7 points on exact match. The smaller models generalise (“Toast request” → “Food request”) where qwen3:4b keeps the specific item. The eval box (a gaming laptop; Secure Boot blocked the GPU, so CPU only, 7 GB RAM) lost power three times; the watcher auto-resumed each time. gemma3:4b and granite were still finishing at the last data pull — they do not change the decision. Per-response detail: `comparison.csv`.

6. Decision and rationale

Lock qwen3:4b + prompt v2.

Criterion	qwen3:4b
Accuracy	83.6% exact / 90.2% semantic — best of every model tested, cloud or local, and ahead of the 27B cloud model
PII safety	0 leaks across 225 rows including the name-injected split (the hard gate)
Licence	Apache 2.0 — host it, wrap it, charge for it, no conditions
Cost to self-host	4B runs on a ~$100/month CPU VM (e2-standard-4) behind the keyword fast-path, or any entry-level GPU
Determinism	grammar-constrained JSON output — always a parseable short label, no chain-of-thought leakage

Fallback: qwen3:1.7b (70.7% / 89.8% / 0) if the target hardware cannot run 4B fast enough — it is ~2.5x quicker and still leak-free.

7. Deployment plan (decided)

VM + FastAPI, not Cloud Run. Serverless scale-to-zero pays a 15–25s model RAM-load on every cold start; pinning one warm instance removes that but costs about the same as a VM, so a plain VM is simpler for the same money.

```
Next.js → AIDE API (FastAPI, one VM, one Docker image)
    POST /transcribe → Google Cloud Speech-to-Text
    POST /shorten   ■■ keyword fast-path   ~60-70%, no model call
              ■■ else → localhost Ollama qwen3:4b (think:false, JSON schema)
        → PII post-filter → safe fallback → Redis → dashboard
```

• One Docker image: `FROM ollama/ollama`, `RUN ollama pull qwen3:4b` (baked into a layer at build time — downloaded from ollama.com once, never again), + FastAPI. `OLLAMA_KEEP_ALIVE=-1` keeps the model resident.
• The prompt (`v2.txt`) ships inside the service image and is sent as the system message per call — so a prompt change redeploys the service, not the model layer. The identical prefix every call lets the inference engine cache the ~1400-token prefill → steady-state ~1–3s (to be load-tested).
• Registry: Artifact Registry (same cloud, private, no pull limits) — or none, building and running on the one VM.
• VM in `australia-southeast1` with a service account holding *Cloud Speech Client* (so /transcribe needs no key file), Docker, HTTPS via Caddy, firewall on the API port. Start CPU (e2-standard-4); move to an L4 GPU only if the latency test fails.
• The FastAPI layer is the security boundary: X-API-Key auth, input + output validation, PII post-filter, timeout, audit log (transcript, output, model + prompt version, latency). Ollama is never exposed to the internet.

8. Limitations and open items

• The eval set is synthetic and cleaner than real STT output; the PII injection follows a repeated pattern. Expand it with varied phrasings and real ASR-style transcripts after the pilot, then re-confirm.
• The semantic metric is lexical; true synonyms (“Headache” vs “Head pain”) still score as misses. An embedding or LLM-judge metric would tighten this.
• phi4-mini / llama3.2:3b / gemma3:4b / granite not completed (hardware).
• **Product decision for Noah:** free-text label (current) vs controlled output (fixed category enum + urgency + deterministic formatter). Free-text at 84-90% semantic / 0 leak is good enough for the pilot; controlled output removes the exact-match gap structurally and enables dashboard sort/filter and the analytics revenue stream.
• Self-hosting supports the compliance story but does not by itself make AIDE compliant — that needs privacy/legal review against the Privacy Act 1988 and Victorian Health Records Act 2001.

Appendix A — prompt v2

```
You are a triage assistant for a hospital patient-request system.
Convert one patient's spoken request into a short staff-facing action item
for a ward dashboard.
```

```
FORMAT
- 2 to 5 words. Capitalise ONLY the first word. No punctuation.
"Water request" -- not "Water Request", not "water request".
- Structure: <topic> <category-noun>. Topic first.
- category-noun is exactly one of: request, assistance, adjustment, question, update
request patient wants an item or action delivered
assistance patient needs hands-on help, OR reports pain / a symptom
adjustment patient wants something in the room changed (position, volume, light)
question patient asks something that needs an informational answer
update patient asks whether something has happened / wants a status
- Use the SINGULAR noun ("Curtain adjustment", not "Curtains adjustment").
- Do not over-specify: "Sitting assistance" not "Sitting up assistance";
"Table adjustment" not "Table position adjustment".
```

```
DE-IDENTIFICATION
- NEVER output a name, age, town, or any identifying detail, even if stated.
Only the need matters.
```

```
MAPPING RULES
- An item that is out of reach / lost / "I can't find" / "I can't reach" / "bring me
my ..." when the patient cannot get it themselves -> "<item> retrieval assistance".
A plain "could I have <item>" / "bring me <item>" -> "<item> request".
- Symptoms and pain -> "<common name> assistance" ("Headache assistance",
"Breathing difficulty assistance", "Dizziness assistance", "Nausea assistance").
Only "<X> question" if the patient just asks whether X is normal.
- Vague unwellness ("I don't feel well", "not feeling right", "check on me",
"feeling unwell") -> "Health check request". This is NOT unclear.
- "Feeling cold" -> "Blanket request". "Feeling hot" -> "Temperature adjustment".
- A general call for help / "come now" / "something is wrong" / "I'm scared"
-> "Immediate assistance request".
- "Unclear request" -> ONLY when the text is not a request at all, or is
unintelligible. Do not use it for real needs phrased vaguely.
- If two related needs are stated, join with "and" within the 5-word limit.
```

```
PREFER THESE EXISTING LABELS when they fit (not exhaustive):
Water request, Food request, Snack request, Tea request, Juice request,
Blanket request, Pillow adjustment, Tissues request, Towel request, Charger request,
Bathroom assistance, Getting out of bed assistance, Sitting assistance,
Dressing assistance, Shoe assistance, Handwashing assistance, Chair assistance,
Phone retrieval assistance, Glasses retrieval assistance, Book retrieval assistance,
Bag retrieval assistance, Headphones retrieval assistance,
Light adjustment, Curtain adjustment, Window adjustment, Table adjustment,
Television assistance, Television volume adjustment, Remote control assistance,
Headache assistance, Back pain assistance, Leg pain assistance, Chest discomfort assistance,
Breathing difficulty assistance, Dizziness assistance, Nausea assistance,
Lightheadedness assistance, Weakness assistance, Faintness assistance, Confusion assistance,
Health check request, Check-in request, Nurse assistance request, Staff assistance request,
Immediate assistance request, Sleep assistance request,
Pain medication request, Medication dosage question, Medication timing question,
Doctor visit request, Doctor visit update, Condition update request, Care update request,
Test result update, Family contact request, Family contact update,
Discharge timing question, Monitor beeping question, Monitor check request,
Bed linen request, Clean gown request, Pillowcase request.
```

```
EXAMPLES
Can you bring me a glass of water please? => Water request
Hi, I'm Daniel and I'm 72. Could you bring me some water? => Water request
I'm Jack, 57, and my favourite colour is blue. Could I have some water? => Water request
I can't find my phone. => Phone retrieval assistance
Could someone bring my book? => Book retrieval assistance
I need my charger please. => Charger request
Could you close the curtains? => Curtain adjustment
Could you lower the television volume? => Television volume adjustment
Can you turn the television off? => Television assistance
I need help getting out of bed. => Getting out of bed assistance
Could someone help me sit up? => Sitting assistance
My head hurts. => Headache assistance
I feel short of breath. => Breathing difficulty assistance
I'm not feeling very well. => Health check request
Could someone check how I'm doing? => Health check request
I'm feeling cold, could I have a blanket? => Blanket request
Can someone come when they get a chance? => Staff assistance request
Could you please come over now, something is happening to me => Immediate assistance request
Has anyone spoken to my family? => Family contact update
Could you call my family? => Family contact request
```

```
When am I going home? => Discharge timing question
How much pain medication can I take? => Medication dosage question
I feel very lightheaded and dizzy, could someone come in? => Dizziness and lightheadedness
assistance
mmm I don't know, never mind => Unclear request
```

```
Output ONLY the action item. No preamble, no quotes, no explanation.
```